

4.1.1 Bipolar Logic Families

The main elements of a bipolar IC are resistors, diodes (which are also capacitors) and transistors. Basically, there are two types of operations in bipolar ICs:

1. Saturated, and
2. Non-saturated.

In saturated logic, the transistors in the IC are driven to saturation, whereas in the case of non-saturated logic, the transistors are not driven into saturation.

The saturated bipolar logic families are:

1. Resistor–transistor logic (RTL),
2. Direct–coupled transistor logic (DCTL),
3. Integrated–injection logic (I²L),
4. Diode–transistor logic (DTL),
5. High–threshold logic (HTL), and
6. Transistor–transistor logic (TTL).

The non-saturated bipolar logic families are:

1. Schottky TTL, and
2. Emitter-coupled logic (ECL).

Table 4.1 *Classification of Digital ICs*

IC Classification	Equivalent individual basic gates	Number of components
Small-scale integration (SSI)	Less than 12	Up to 99
Medium-scale integration (MSI)	12–99	100–999
Large-scale integration (LSI)	100–999	1,000–9,999
Very large-scale integration (VLSI)	Above 1,000	Above 10,000

The various characteristics of digital ICs used to compare their performances are:

1. Speed of operation,
2. Power dissipation,
3. Figure of merit,
4. Fan-out,
5. Current and voltage parameters,
6. Noise immunity,
7. Operating temperature range,
8. Power supply requirements, and
9. Flexibilities available.

4.2.1 Speed of Operation

The speed of a digital circuit is specified in terms of the propagation delay time. The input and output waveforms of a logic gate are shown in Fig. 4.1. The delay times are measured between the 50 per cent voltage levels of

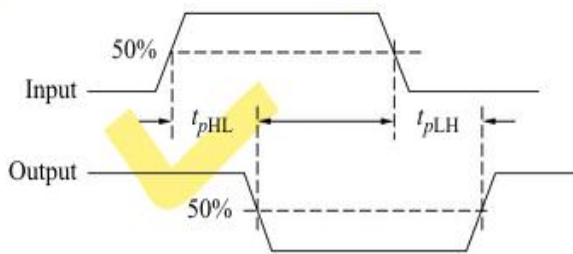


Fig. 4.1 Input and Output Voltage Waveforms to Define Propagation Delay Times

input and output waveforms. There are two delay times: t_{pHL} , when the output goes from the HIGH state to the LOW state and t_{pLH} , corresponding to the output making a transition from the LOW state to the HIGH state. The propagation delay time of the logic gate is taken as the average of these two delay times.

4.2.2 Power Dissipation

This is the amount of power dissipated in an IC. It is determined by the current, I_{CC} , that it draws from the V_{CC} supply, and is given by $V_{CC} \times I_{CC}$. I_{CC} is the average value of $I_{CC}(0)$ and $I_{CC}(1)$. This power is specified in milliwatts. It is known as static power dissipation, i.e., the power consumed by the circuit when input signals are not changing.

4.2.3 Figure of Merit

The figure of merit of a digital IC is defined as the product of speed and power. The speed is specified in terms of propagation delay time expressed in nanoseconds. by

$$\text{Figure of merit} = \text{propagation delay time (ns)} \times \text{power (mW)}$$

It is specified in pico joules ($\text{ns} \times \text{mW} = \text{pJ}$)

A low value of speed-power product is desirable. In a digital circuit, if it is desired to have high speed, i.e. low propagation delay, then there is a corresponding increase in the power dissipation and vice-versa.

✓ Power Dissipation: is the amount of power dissipated in an IC in milli watts in the form of heat.

$P_D = V_{CC} \times I_{CC}$

- excessive heat/temp. can damage the IC, So, power dissipation should be less
- determined by the current I_{CC} that is drawn from V_{CC}

V_{CC} for TTL IC
 V_{DD} for CMOS IC

$$I_{CC}(\text{Avg}) = \frac{I_{CCH} + I_{CCL}}{2}$$

$P_D = V_{CC} \times I_{CC}$

I_{CCH} = current drawn from the power supply when o/p voltage is at high level

I_{CCL} = at low level

$$P_D(\text{Avg}) = V_{CC} \times I_{CC}(\text{Avg})$$

$$= V_{CC} \times \frac{I_{CCH} + I_{CCL}}{2}$$

4.2.4 Fan-Out

the maximum logic inputs (of the same logic families) that an output can drive reliably.

This is the number of similar gates which can be driven by a gate. High fan-out is advantageous because it reduces the need for additional drivers to drive more gates.

4.2.5 Current and Voltage Parameters

The following currents and voltages are specified which are very useful in the design of digital systems.

High-level input voltage, V_{IH} : This is the minimum input voltage which is recognised by the gate as logic 1.

Low-level input voltage, V_{IL} : This is the maximum input voltage which is recognised by the gate as logic 0.

$$= \min\left(\frac{I_{OH}}{I_{IH}}\right)$$

4.2.6 Noise Immunity

The input and output voltage levels defined above are shown in Fig. 4.3. Stray electric and magnetic fields may induce unwanted voltages, known as *noise*, on the connecting wires between logic circuits. This may

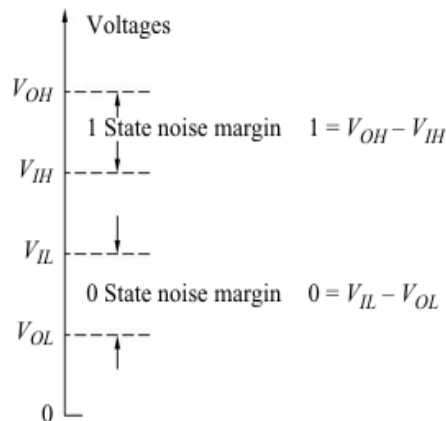
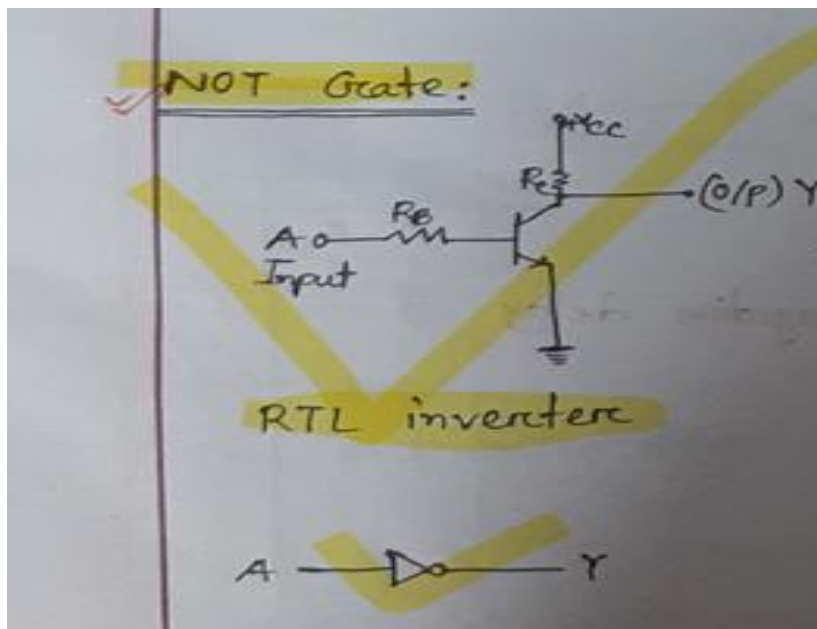


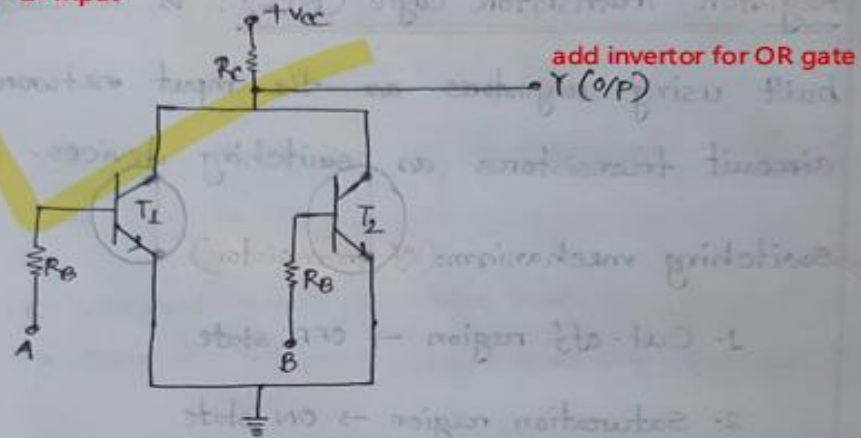
Fig. 4.3 Voltage Levels and Noise Margins of ICs

cause the voltage at the input to a logic circuit to drop below V_{IH} or rise above V_{IL} and may produce undesired operation. The circuit's ability to tolerate noise signals is referred to as the *noise immunity*, a quantitative measure of which is called *noise margin*. Noise margins are illustrated in Fig. 4.3.

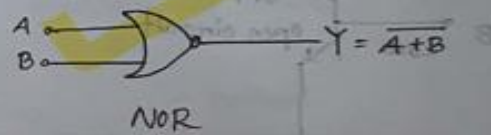
The noise margins defined above are referred to as *dc noise margins*. Strictly speaking, the noise is generally thought of as an a.c. signal with amplitude and pulse width. For high speed ICs, a pulse width of a few microseconds is extremely long in comparison to the propagation delay time of the circuit and therefore, may be treated as d.c. as far as the response of the logic circuit is concerned. As the noise pulse width decreases and



NOR Gate: 2-input



A	B	Y
Low	Low	High
Low	High	Low
High	Low	Low
High	High	Low



* 3-Input RTL NOR gate with working principle.

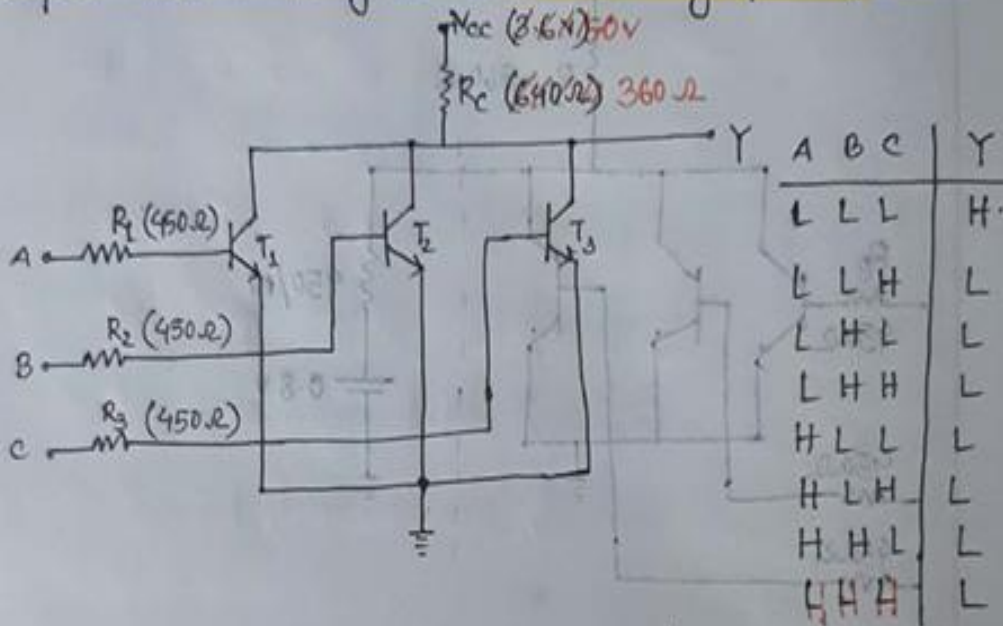


Figure: 3-input RTL NOR gate

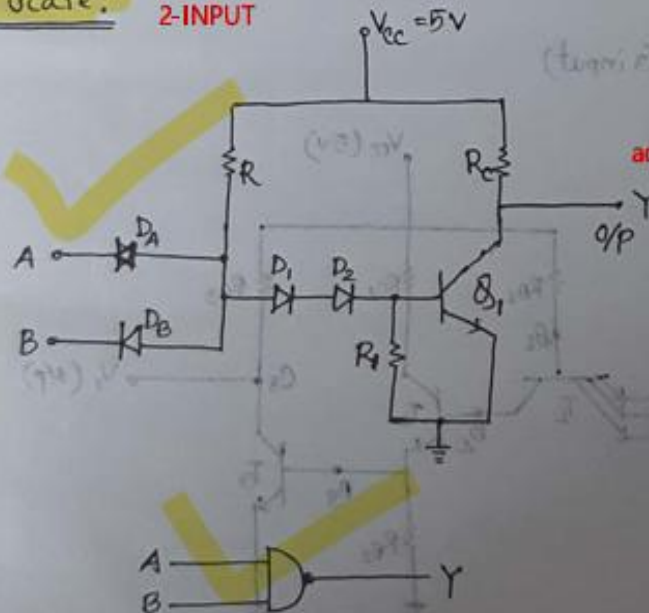
The figure above shows that T_1, T_2, T_3 are three transistors and A, B, C are the three inputs, given to the base of the three transistors. Y is the output.

When all the three inputs are low or 0, if the transistors are in the cut-off region. The collector and emitter terminal will act as an open circuit and the V_{cc} will pass through the output, making the output high or 1.

When any of the ~~trans~~ input is high or 1, the transistor will be in its saturated region and will act as the e and c terminal will act as short circuit. So, the current will flow it will make a path for the supply voltage V_{cc} to go to the ground through the resistor R_c and transistor. Thus there will be 0 or low at o/p terminal Y .

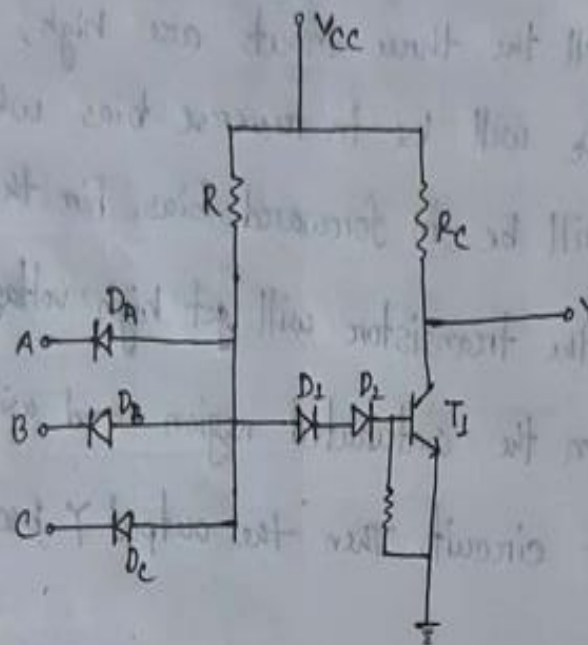
MATH:

NAND Gate: 2-INPUT



A	B	Y
Low	Low	High
Low	High	High
High	Low	High
High	High	Low

3-input DTL NAND Gate and its working principle



A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Figure- 3-input DTL NAND gate

When the inputs are low signal or 0, the D_A , and D_B , D_C diodes will act as forward bias, whereas D_1 and D_2 will act as reverse bias. As D_1 and D_2 are working as reverse bias, the base current of transistor T_1 will be low, means the transistor will be in the cut off region and will act as an open circuit. So the output will be V_{CC} which will pass through the R_C .

Again, when one of the input will be high or 1, then one of the three diodes will be forward biased and other will be in reverse biased. So then D_1 and D_2 will be in the reverse bias. As a result the base of the transistor will get low voltage or 0. Then the transistor will be in the cut off region.

it will be in the saturated region and will work as a short circuit. Then the output Y becomes low or 0 voltage.

3-input NAND TTL NAND and its working principle.

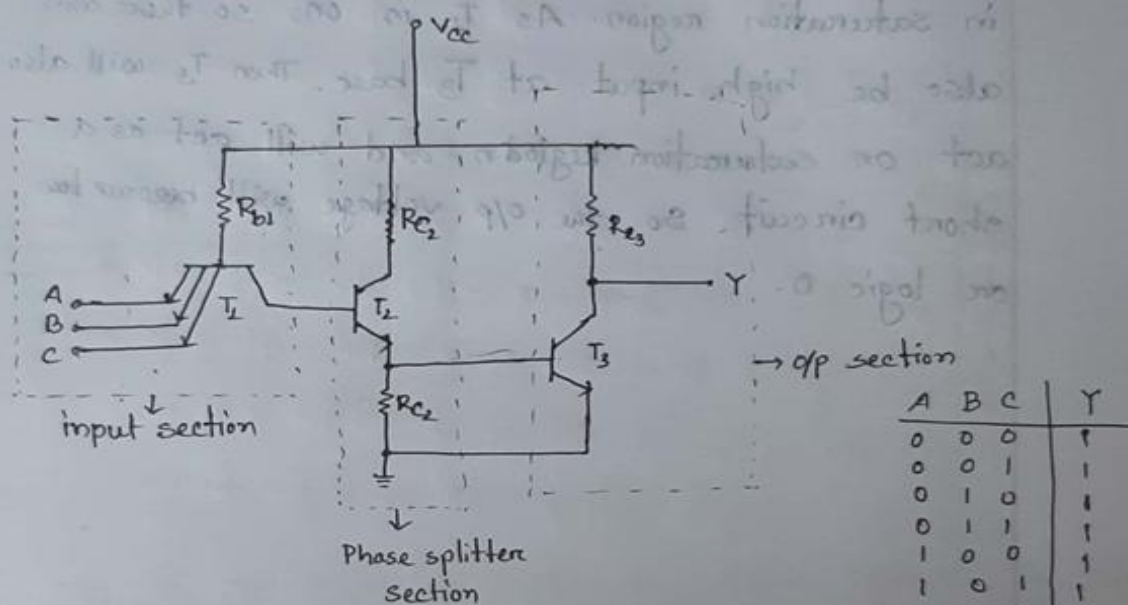


Figure: 3-input TTL NAND

When any of the input is low means logic 0, the base-emitter for T_1 transistor will act as forward bias and base-collector will work as reverse bias. For this reason, base input for T_2 will be low, so it will be on cut off region. As T_2 is in cut off region, T_3 will also be on cut-off region. So, the output voltage will be high or 1.

On the other hand, when the inputs are all 1 or high, the base-emitter for T_1 will work as reverse bias and base collector will work as forward bias. For this reason, base

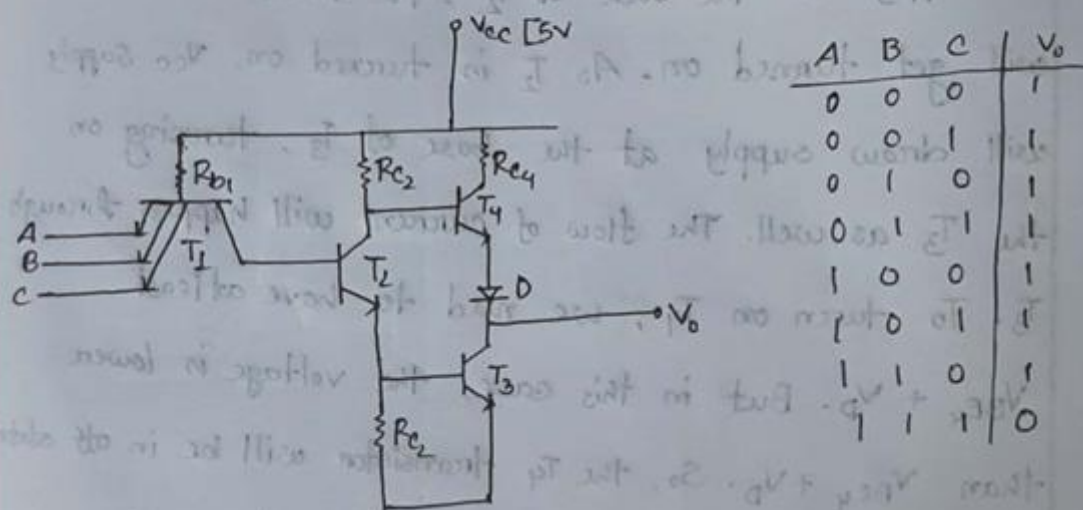
input for T_2 will become high and T_2 will act in saturation region. As T_2 is on, so there will also be high input at T_3 base. Then T_3 will also act on saturation region and will act as a short circuit. So, the o/p voltage will become low or logic 0.

4.8.2 Current Sink Logic

When at least one of the inputs is LOW, current I_{B1} from V_{CC} source flows through the input signal.

$$I_{B1} = \frac{V_{CC} - V_{B1}}{R_{B1}} = \frac{5 - 0.9}{4} \text{ mA} = 1.025 \text{ mA}$$

* TTL NAND Gate with Totem pole Output



If any of the input is active low, means logic 0, In that case, the base-emitter junction of T_1 will be forward biased and base-collector junction will be in reverse bias. T_1 will be in the active region and open circuited at base collector junction. So, no input will be passed on base of T_2 , so T_2 will be off and open circuited. So, this will not allow supply to be given at the base of T_3 . Eventually T_3 will be also in off state. Supply voltage will be passed on the base of T_4 and T_4 will be in on state and will be short circuited. So, to calculate this o/p voltage, $V_o = V_{cc} - I_C R_{C4}(\text{sat}) - V_D$.

When all the inputs are provided with logic 1, the base-emitter junction will be in reverse bias and the base collector junction

→ The limitation of RTL & DTL circuits:

2-input CMOS OR gate

DTL

① Low operating speed: around 1MHz

② High power consumption.

③ fan-out Limitation.

④ Temperature sensitivity.

⑤ slower switching.

RTL:

① High power Dissipation

② Low noise margin

③ Low component density

④ Limited high-speed performance

⑤ High Switching current

RTL:

① Poor noise immunity.

② Low Fan-out.

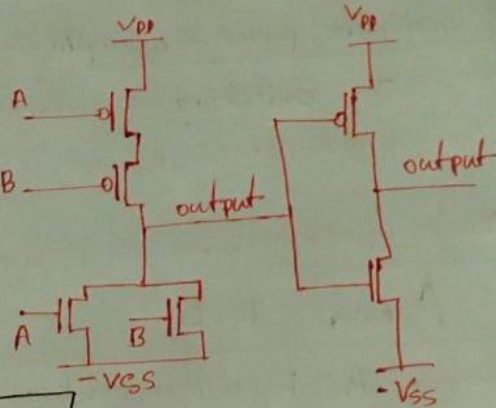
③ Low Switching speed.

④ High power Dissipation.

⑤ obsolete.

why TTL is most popular?

Because it provides a good compromise between speed, power consumption, cost and reliability, along with easy availability and wide range of ICs.



① Fast switching speeds

② simple interfacing

③ High drive capability

④ good fan-out

⑤ moderate power consumption.

Chapter 4

Modern Digital Electronics - R. P. Jain
4th

Discuss the switching characteristics of a semiconductor diode.

Refer APV note: 26 page

→ The comparison between DTL & TTL, RTL

features	RTL	DTL	TTL
components	Resistor and transistor	Diodes and transistor	Only transistor
speed	Slowest	Medium (Fast than RTL)	Fastest
Fan-out	Low (3-5)	moderate (approx 10)	High (approx 20)
Noise margin	Low	moderate	high/good
power dissipation	High	Low/moderate	Low to high
complexity	simplest design	moderate design	more complex
primary logic	NOR gate	NAND gate	NAND gate

Switching Property:

A semiconductor diode switches between a low-resistance "ON" state (forward bias) and a high-resistance "OFF" state (reverse bias)

Diode

(Anode) P ———— N (Cathode)

F.B → Switch on

R.B → Switch off

Forward Bias (ON state): When a forward voltage is applied, the diode acts as a closed switch.

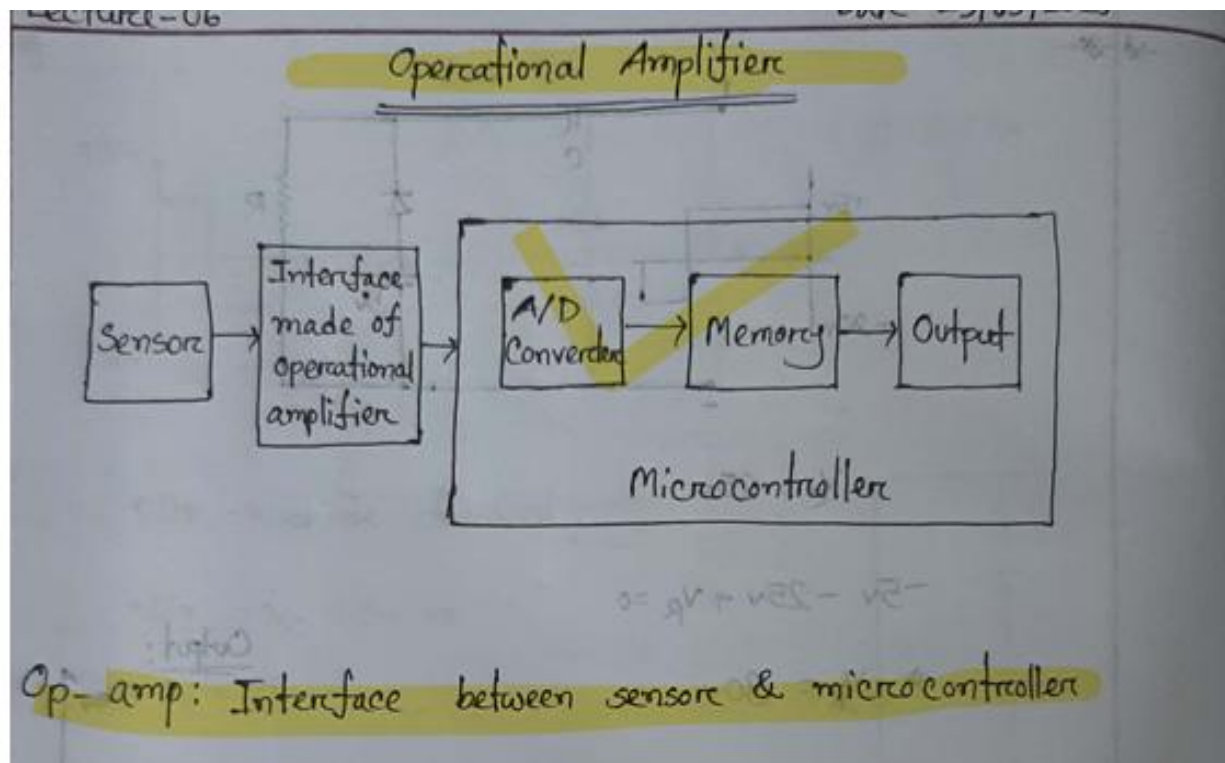
Reverse Bias (OFF state): When polarity reverses, the diode stops conducting, acting as an open switch.

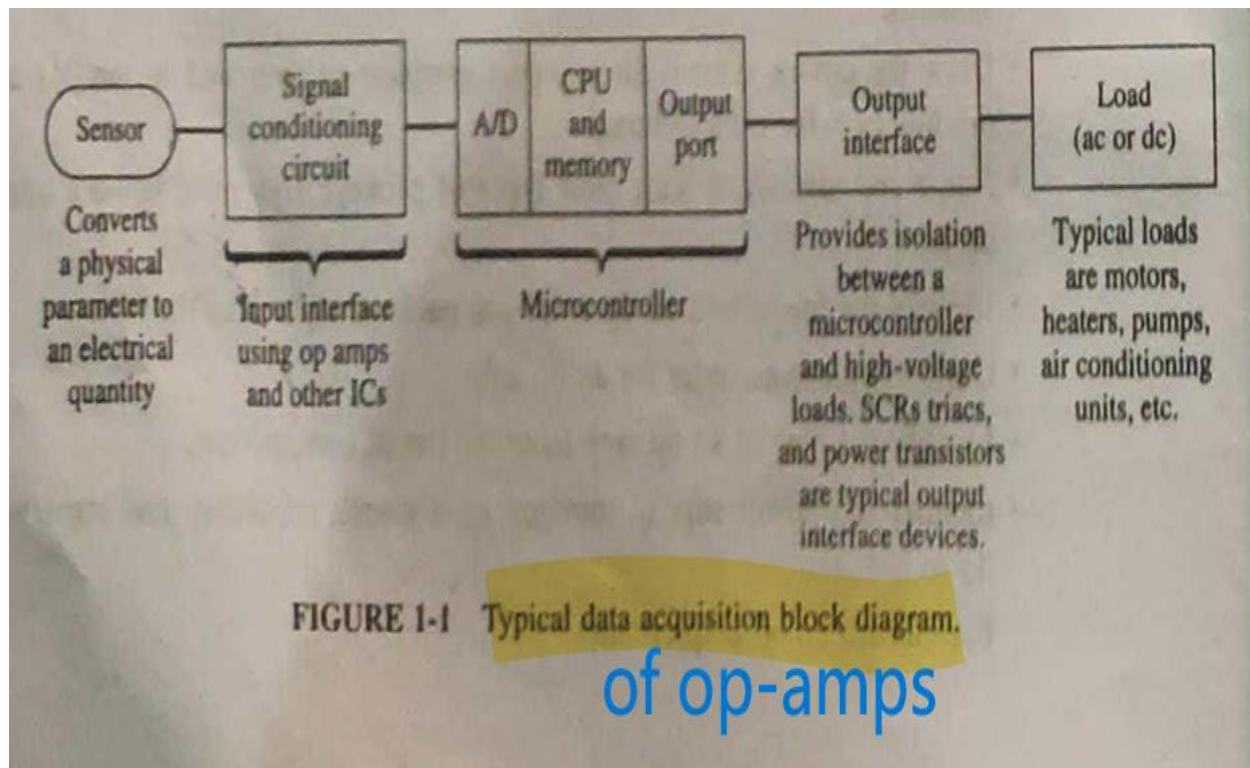
Transistor

c (Collector)

Switch on

Switch off





Characteristics of an Ideal Op-Amp

- Infinite input impedance
- Zero output impedance
- Zero common-mode gain, or, infinite common-mode rejection
- Infinite open-loop gain A
- Infinite bandwidth.

1-2.2 Simplified Internal Circuitry of a General-Purpose Op Amp

General-purpose op amps are multistage systems. As shown in Fig. 1-3(a), the basic op amp consists of an input stage with two input terminals, an output stage with one output terminal, and an intermediate stage that connects the output signal of the input stage to the input terminal of the output stage.

Dc power is applied from a bipolar supply to the op amp's external power supply terminals and thus to each internal stage of the op amp. Depending on the application, input signals, $V_{(+)}$ and $V_{(-)}$ can be positive, negative, or zero. The resulting output voltage is measured across the load resistor R_L , which is connected between the op amp's output

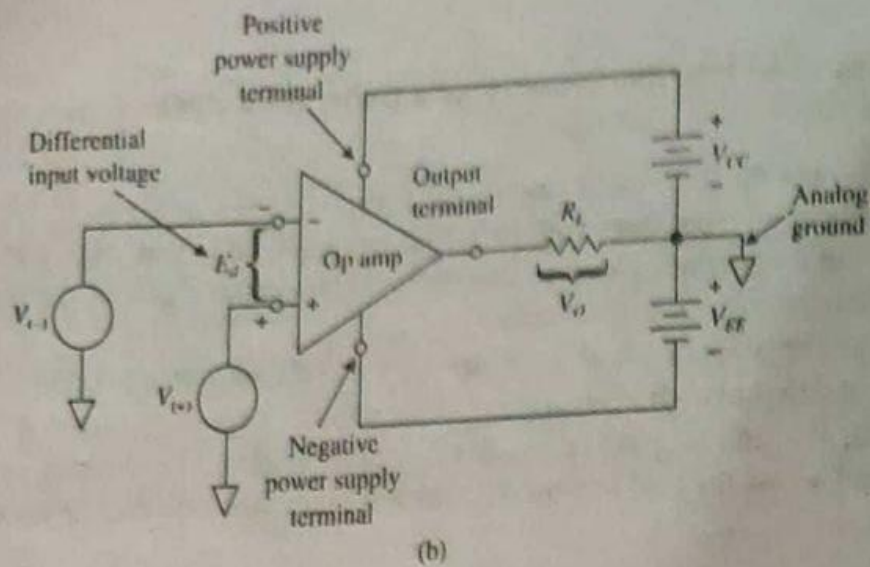


FIGURE 1-3(b)

terminal and common. The output voltage, V_o , depends on the input signals and characteristics of the op amp.

The ideal op amp of Fig. 2-1 has infinite gain and infinite frequency response. The input terminals draw no signal or bias currents and exhibit infinite input resistance. Output impedance is zero ohms, and the power supply voltages are without limit. We now examine the function of each op amp terminal to learn something about the limitations of a real op amp.

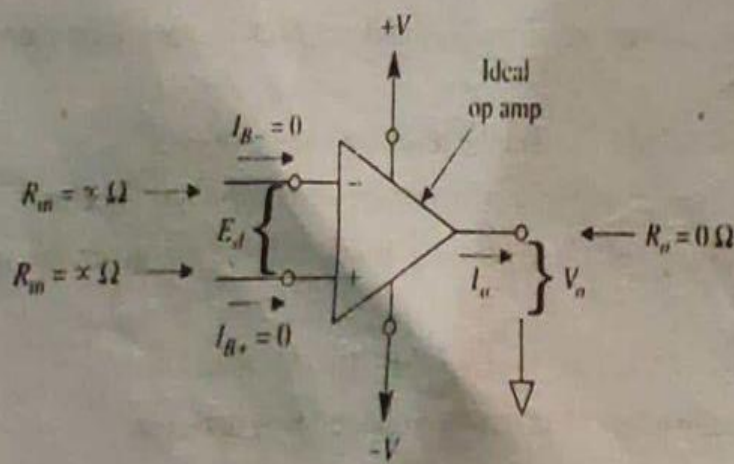


FIGURE 2-1 The ideal op amp has infinite gain and input resistances plus zero output resistance.

2-2.1 Definition

Refer to Fig. 2-3. Output voltage V_o is determined by E_d and the *open-loop voltage gain*, A_{OL} . A_{OL} is called open-loop voltage gain because possible feedback connections from output terminal to input terminals are left open. Accordingly, V_o is expressed by the relationship

$$\text{output voltage} = \text{differential input voltage} \times \text{open-loop gain} \quad (2-2)$$

$$V_o = E_d \times A_{OL}$$

2-2.2 Differential Input Voltage, E_d

The value of A_{OL} is extremely large, often 200,000 or more. Recall from Section 2-1.2 that V_o can never exceed the positive or negative saturation voltages $+V_{sat}$ and $-V_{sat}$. For a $\pm 15\text{-V}$ supply, the saturation voltages are approximately $\pm 13\text{ V}$. Thus, for the op amp to act as an amplifier, E_d must be limited to a maximum voltage of $\pm 65\text{ }\mu\text{V}$. This conclusion is reached by rearranging Eq. (2-2).

$$E_{d\text{ max}} = \frac{+V_{sat}}{A_{OL}} = \frac{13\text{ V}}{200,000} = 65\text{ }\mu\text{V}$$

$$-E_{d\text{ max}} = \frac{-V_{sat}}{A_{OL}} = \frac{-13\text{ V}}{200,000} = -65\text{ }\mu\text{V}$$

In the laboratory or shop it is difficult to measure $65\text{ }\mu\text{V}$, because induced noise, 60-Hz hum, and leakage currents on the typical test setup can easily generate a millivolt ($1000\text{ }\mu\text{V}$). Furthermore, it is difficult and inconvenient to measure very high gains. The op amp also has tiny internal unbalances that act as a small voltage that may exceed E_d . As mentioned in Section 2-1.4, this small voltage is called an *offset voltage* and is discussed in Chapter 9.

3-1.4 Negative Voltage Applied to the Inverting Input

Figure 3-2 shows a negative voltage, E_i , applied via R_i to the inverting input. All the principles and equations of Sections 3-1.1 to 3-1.3 still apply. The only difference between Figs. 3-1 and 3-2 is the direction of the currents. Reversing the polarity of the input voltage, E_i , reverses the direction of all currents and the voltage polarities. Now the output of the amplifier will go positive when E_i goes negative.

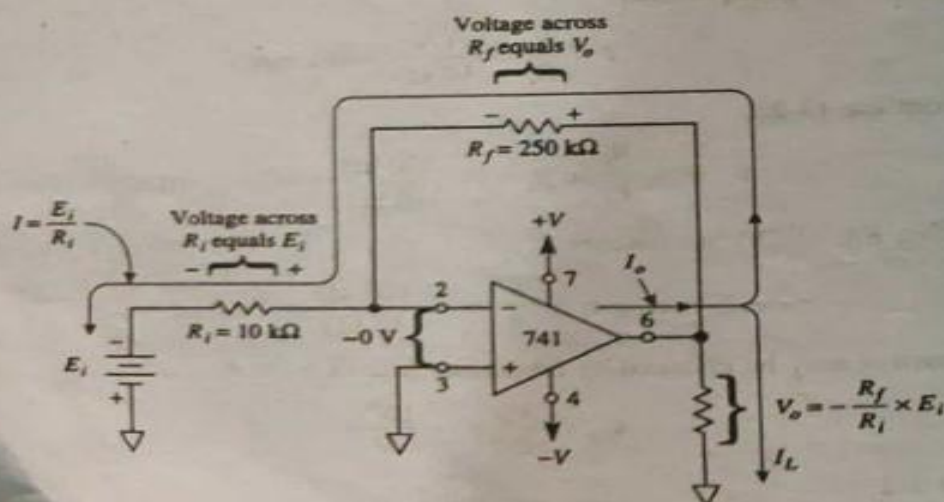


FIGURE 3-2 Negative voltage applied to the (–) input of an inverting amplifier.

Example 3-3

For Fig. 3-2, let $R_f = 250 \text{ k}\Omega$, $R_i = 10 \text{ k}\Omega$, and $E_i = -0.5 \text{ V}$. Calculate (a) I ; (b) the voltage across R_f ; (c) V_o .

Solution (a) From Eq. (3-1a),

$$I = \frac{E_i}{R_i} = \frac{0.5 \text{ V}}{10 \text{ k}\Omega} = 50 \text{ }\mu\text{A} = 0.05 \text{ mA}$$

(b) From Eq. (3-1b),

$$\begin{aligned} V_{R_f} &= I \times R_f \\ &= (50 \text{ }\mu\text{A})(250 \text{ k}\Omega) \\ &= 12.5 \text{ V} \end{aligned}$$

(c) From Eq. (3-2a),

$$V_o = -\frac{R_f}{R_i} \times E_i = -\frac{250 \text{ k}\Omega}{10 \text{ k}\Omega} (-0.5 \text{ V}) = +12.5 \text{ V}$$

Thus the magnitude of the output voltage does equal the voltage across R_f , and $A_{CL} = -25$.

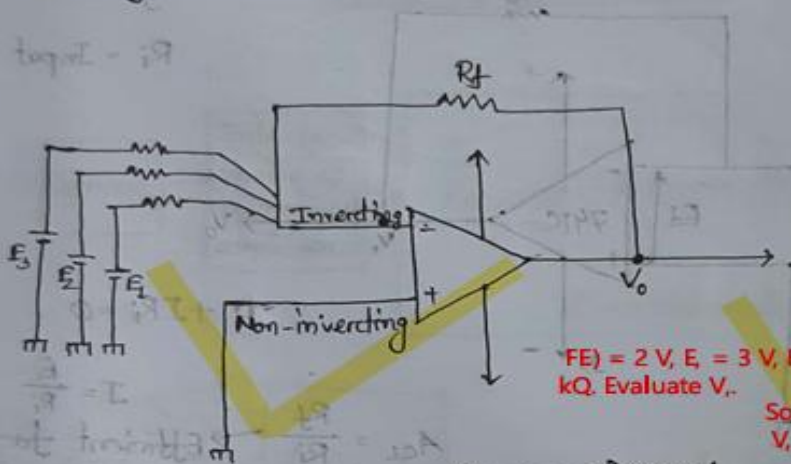
Example 3-5

For the circuit of Fig. 3-3, $R_f = 20 \text{ k}\Omega$ and $R_i = 10 \text{ k}\Omega$, calculate the voltage gain A_{CL} .

Solution From Eq. (3-2b),

$$A_{CL} = -\frac{R_f}{R_i} = -\frac{20 \text{ k}\Omega}{10 \text{ k}\Omega} = -2$$

Inverting Adder



$E_1 = 2 \text{ V}$, $E_2 = 3 \text{ V}$, $E_3 = 1 \text{ V}$, and all resistors are $10 \text{ k}\Omega$. Evaluate V_o .

Solution:

$$V_o = -(2\text{V} + 3\text{V} + 1\text{V}) = -6\text{V}$$

$$V_{R_f} = (I_1 + I_2 + I_3) \times R_f$$

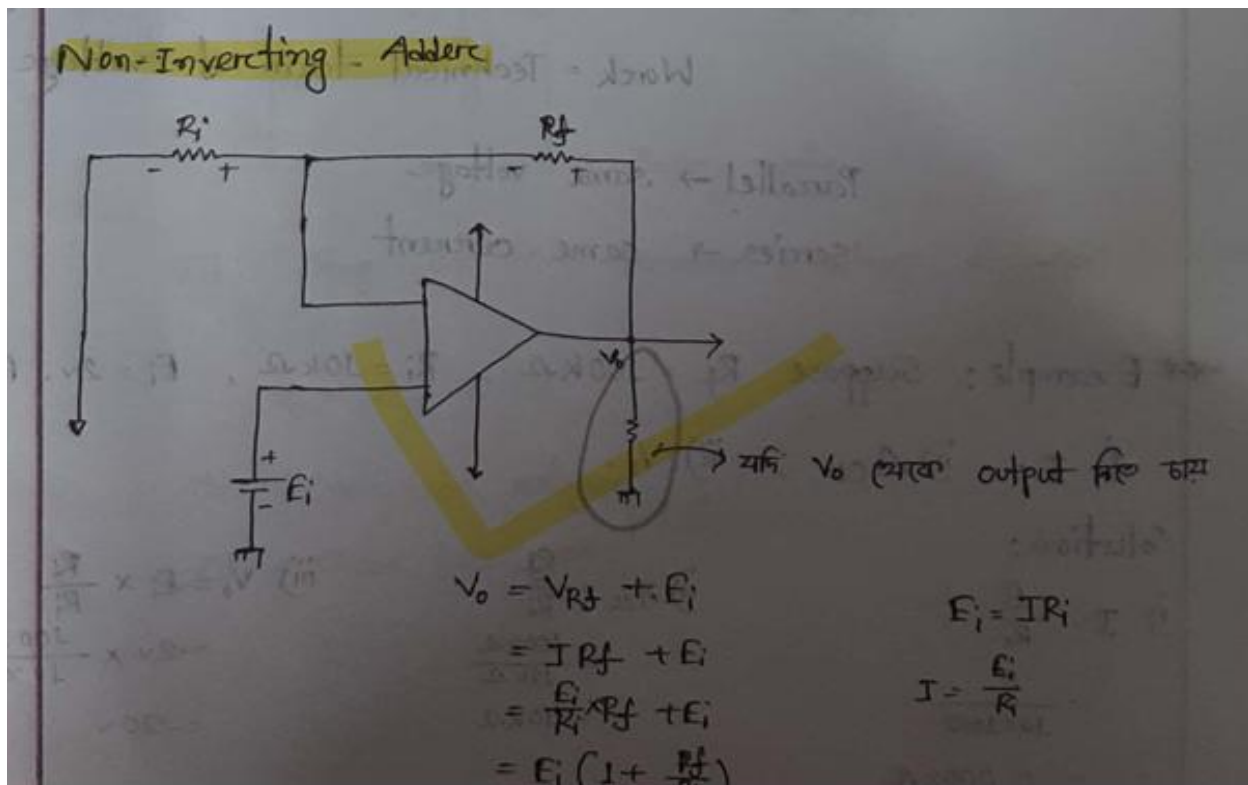
$$V_o = -(I_1 + I_2 + I_3) \times R_f$$

$$= -\left(\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}\right) \times R_f$$

⚙️ Working Principle:

- The inverting terminal is at **virtual ground (0 V)** due to negative feedback.
- Each input voltage produces a current through its resistor.
- All currents are **added together at the inverting node**.
- The op-amp converts this total current into a single output voltage.

Solution From Eq. (3-4), $V_n = -(2\text{ V} + 3\text{ V} + 1\text{ V}) = -6\text{ V}$. The PSpice model for this circuit is shown in Fig. 3-24.



⚙️ Working Principle:

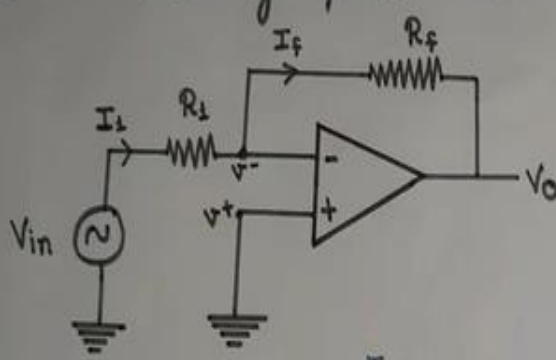
- Inputs are first combined using a **resistor network (voltage averaging/summing)**.
- The resulting voltage is applied to the **non-inverting input**.
- Op-amp amplifies this combined voltage **without phase inversion**.
- Output follows input in same polarity.

Explain how an operational amplifier can be used to design / how voltage gain is obtained for
 , Inverting and non-inverting amplifiers

- (a). What is an operational amplifier? "The inverting amplifier provides a constant voltage gain with 180° phase shift" - justify the statement with relevant circuit diagram.

Inverting Amplifier :-

An inverting amplifier is a type of operational amplifier circuit which produces an output which is out of phase with respect to its input by 180°.



$$I_1 = I_f$$

$$\frac{V_{in} - v^-}{R_1} = \frac{v^- - V_o}{R_f}$$

$v^+ = 0$ (grounded) by virtual ground concept;
 $v^- = v^+ = 0$

$$\frac{V_{in}}{R_1} = \frac{-V_o}{R_f}$$

$$V_o = -\frac{R_f}{R_1} \times V_{in}$$

$$\frac{V_o}{V_{in}} = -\frac{R_f}{R_1}$$

✓ Working Principle:

- The input signal is applied to the **inverting (-) terminal** of the op-amp.
- The **non-inverting (+) terminal** is grounded.
- Due to very high gain, the op-amp creates a **virtual ground** at the inverting terminal (voltage $\approx 0V$).
- Current flows through the input resistor (R_1) and feedback resistor (R_f).
- The output adjusts itself to maintain the virtual ground condition.

✓ Key Result:



- The output signal is **180° out of phase (inverted)** with the input.

- **Closed-loop gain:**

$$A_{cl} = -\frac{R_f}{R_1}$$

- **Current:**

$$I = \frac{V_{in}}{R_1}$$

$$V_o = -E_i \frac{R_f}{R_i} \quad (3-2a)$$

Now, introducing the definition that the closed-loop gain of the amplifier is A_{CL} , we rewrite Eq. (3-2a) as

$$A_{CL} = \frac{V_o}{E_i} = -\frac{R_f}{R_i} \quad (3-2b)$$

Example 3-1

For Fig. 3-1, let $R_f = 100 \text{ k}\Omega$, $R_i = 10 \text{ k}\Omega$, and $E_i = 1 \text{ V}$. Calculate (a) I ; (b) V_o ; (c) A_{CL} .

Solution (a) From Eq. (3-1a),

$$I = \frac{E_i}{R_i} = \frac{1 \text{ V}}{10 \text{ k}\Omega} = 0.1 \text{ mA}$$

(b) From Eq. (3-2a),

$$V_o = -\frac{R_f}{R_i} \times E_i = -\frac{100 \text{ k}\Omega}{10 \text{ k}\Omega}(1 \text{ V}) = -10 \text{ V}$$

(c) Using Eq. (3-2b), we obtain

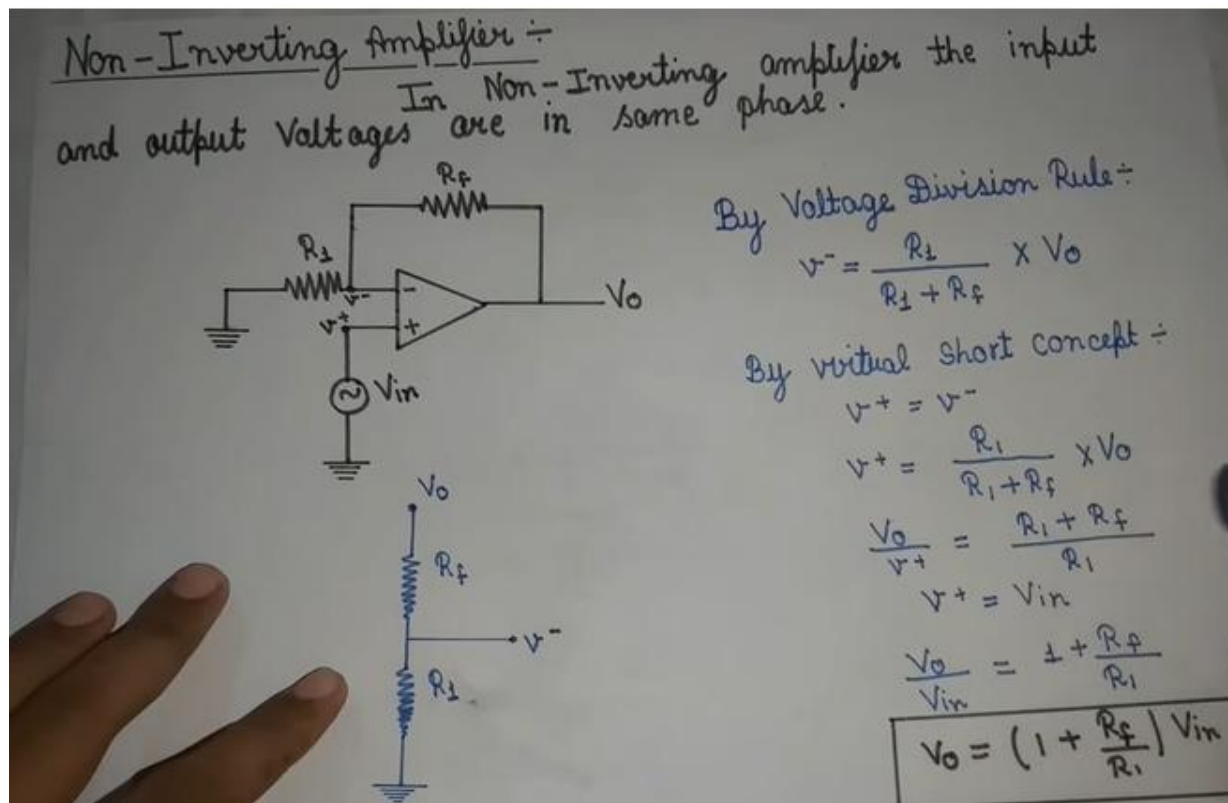
$$A_{CL} = -\frac{R_f}{R_i} = -\frac{100 \text{ k}\Omega}{10 \text{ k}\Omega} = -10$$

This answer may be checked by taking the ratio of V_o to E_i :

$$A_{CL} = \frac{V_o}{E_i} = \frac{-10 \text{ V}}{1 \text{ V}} = -10$$

Example 3-2

Using the values given in Example 3-1 and $R_L = 25 \text{ k}\Omega$, determine (a) I_L ; (b) the total current into the output pin of the op amp.



✓ Working Principle:

- The input signal is applied to the **non-inverting (+) terminal**.
- The inverting (-) terminal is connected to a **voltage divider** (R_1 and R_f) from output to ground.
- The op-amp adjusts the output so that the voltage at both terminals becomes equal ($V^+ = V^-$).
- Feedback ensures stable amplification.

✓ Key Result:

- The output signal is **in phase (same phase)** as the input.

Example 3-13

(a) Find the voltage gain for the noninverting amplifier of Fig. 3-8. If E_i is a 100-Hz triangle wave with a 2-V peak, plot (b) V_o vs. t ; (c) V_o vs. E_i .

Solution (a) From Eq. (3-10b),

$$A_{CL} = \frac{R_f + R_1}{R_1} = \frac{(40 + 10) \text{ k}\Omega}{10 \text{ k}\Omega} = 5$$

3-6 VOLTAGE FOLLOWER

3-6.1 Introduction

The circuit of Fig. 3-9 is called a voltage follower, but it is also referred to as a source follower, unity-gain amplifier, buffer amplifier, or isolation amplifier. It is a special case of the noninverting amplifier. The input voltage, E_i , is applied directly to the (+) input. Since the voltage between (+) and (-) pins of the op amp can be considered 0,

$$V_o = E_i \quad (3-11a)$$

Note that the output voltage equals the input voltage in both magnitude and sign. Therefore, as the name of the circuit implies, the output voltage *follows* the input or source voltage. The voltage gain is 1 (or unity), as shown by

$$A_{CL} = \frac{V_o}{E_i} = 1 \quad (3-11b)$$

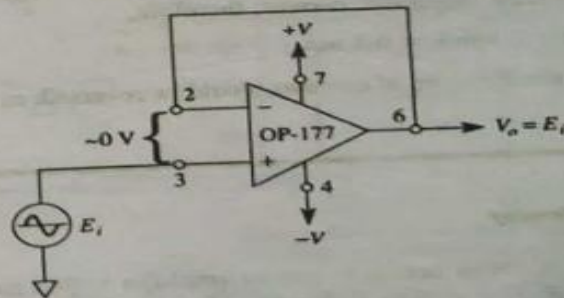


FIGURE 3-9 Voltage follower.

Date - 02/07/2023

Subtractor

$S_1, S_2 = \text{input}$

$$V_1 = -S_2 \cdot \frac{R_f}{R_i}$$

$$= -S_2 \cdot \frac{10k\Omega}{10k\Omega}$$

$$= -S_2$$

$$-S_2 + S_1$$

Non-inverting, $A_{CL} = 1$

$$A_{CL} = \frac{V_o}{E_i} = 1 + \frac{R_f}{R_i}$$

$$= 1 + \frac{0}{R_i}$$

$$= 1 + 0$$

$$= 1$$

$$V_o = E_i \left(1 + \frac{R_f}{R_i}\right)$$

✓ Working Principle:

The input signal is applied to the inverting (–) terminal through a resistor.

A capacitor is used in the feedback path instead of a resistor.

The op-amp continuously sums (integrates) the input signal over time.

The non-inverting (+) terminal is grounded.

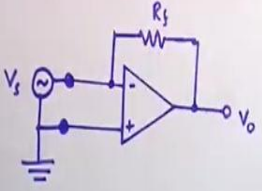
✓ Key Result:

The output is proportional to the integral of the input signal and is inverted.

The difference between close loop and open loop Gain

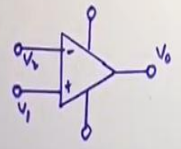
Closed Loop Voltage Gain :-

- Closed Loop voltage gain is the voltage gain of an operational amplifier with feedback.
- Closed Loop voltage gain of an OPAMP is defined as the ratio of output voltage to input signal voltage.


$$A_{CL} = \frac{V_o}{V_s}$$

Open Loop Voltage Gain :-

- It is the voltage Gain of an OP AMP without feedback.
- It is the maximum possible voltage gain from an Operation amplifier.
- It is defined as the ratio of output voltage to the differential input voltage.


$$A_{OL} = \frac{V_o}{V_1 - V_2}$$
$$A_{OL} = \frac{V_o}{V_d}$$

Why op-amps are used?

Operational amplifiers (op-amps) are used as versatile, low-cost building blocks in analog electronics to amplify, filter, and condition signals due to their high gain, high input impedance, and low output impedance. They simplify circuit design for tasks like signal boosting, voltage regulation, and analog-to-digital conversion

Showing the number of pin of 741C op-amp.

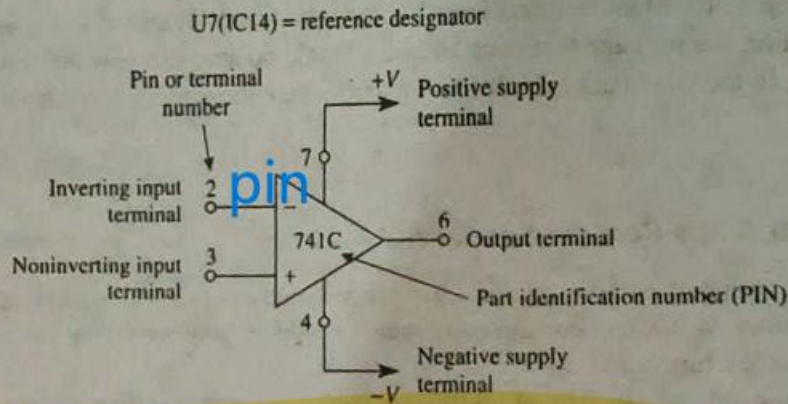


FIGURE 1-2 Circuit symbol for the general-purpose op amp. Pin numbering is for an 8-pin mini-DIP package.

a *part identification number* (PIN) placed within the triangular symbol. The PIN refers to a particular op amp with specific characteristics. The 741C op amp illustrated here is a general-purpose op amp that is used throughout the book for illustrative purposes.

The op amp may also be coded on a circuit schematic with a *reference designator* such as U7, IC14, and so on. Its PIN is then placed beside the reference designator in the parts list of the circuit schematic. All op amps have at least five terminals: (1) The positive power supply terminal V_{CC} or $+V$ at pin 7, (2) the negative power supply terminal V_{EE} or $-V$ at pin 4, (3) output pin 6, (4) the inverting ($-$) input terminal at pin 2, and (5) the noninvert-

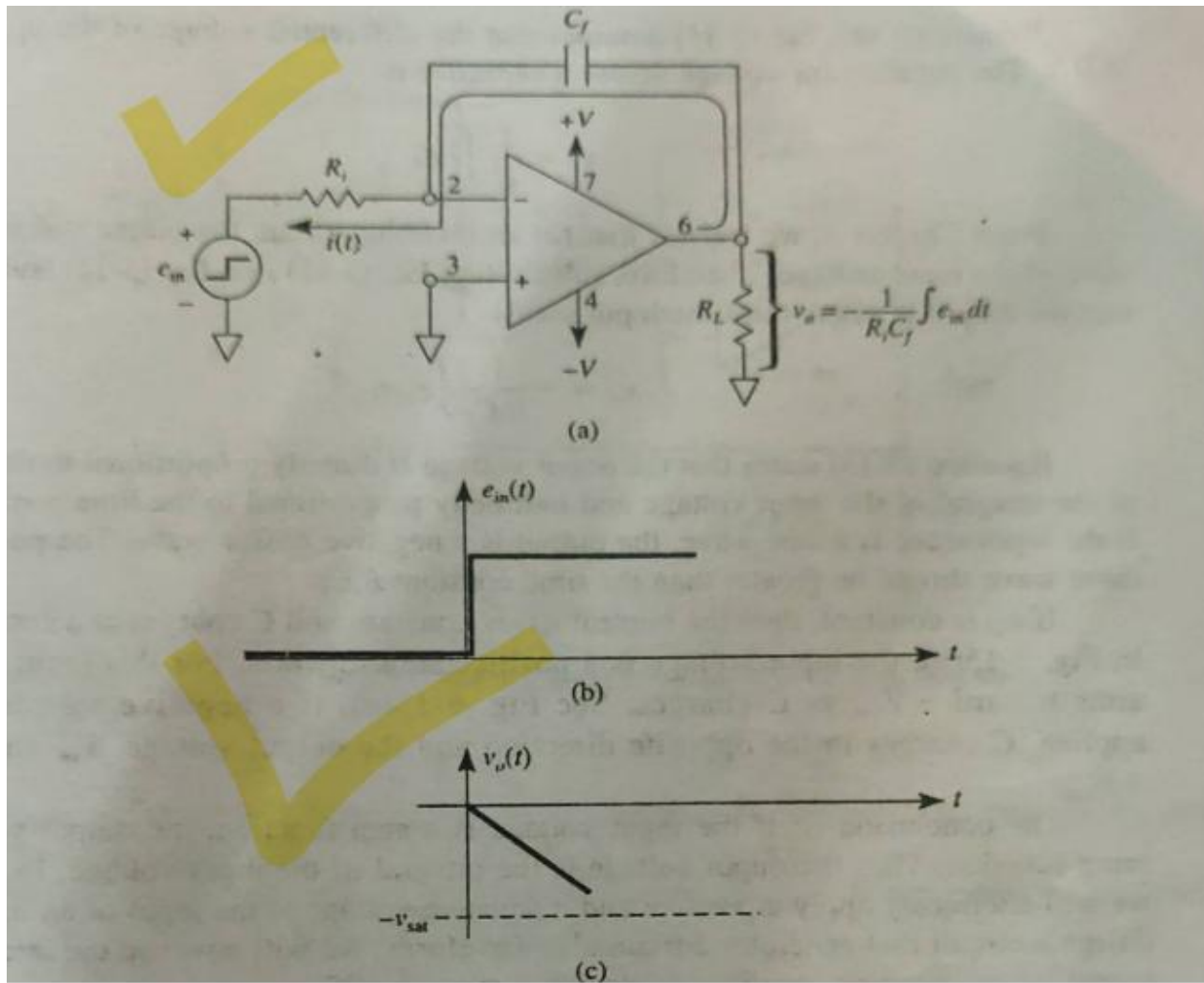
Explain how an operational amplifier can be used to design an integrator.

FIGURE 5-15 (a) An integrator circuit. The output voltage ramps negative in (c) for a positive input step function in (b).

5-12.1 Integrators

If the feedback resistor in an inverting amplifier is replaced by a capacitor as shown in Fig. 5.15(a), the new op amp circuit is known as an integrator. Integrators are used in the design of signal generators and signal processing circuits. Chapter 6 describes signal generating circuits, but the basics of an integrator and a signal processing application are described here. In Fig. 5-15(a), the current that charges the capacitor is set by e_{in} and R_i and is given by

$$i(t) = \frac{e_{in}}{R_i} \quad (5-11)$$



✓ Working Principle:

The input signal is applied to the inverting (–) terminal through a resistor.

A capacitor is used in the feedback path instead of a resistor.

The non-inverting (+) terminal is grounded.

The op-amp continuously accumulates the input signal over time, producing an output based on integration.

✓ Key Result:



The output is proportional to the integral of the input signal and is inverted (180° phase shift).

Remember that Eq. (5-11) assumes that the differential voltage of the op amp, E_{db} , is 0 V. The equation for voltage across a capacitor is

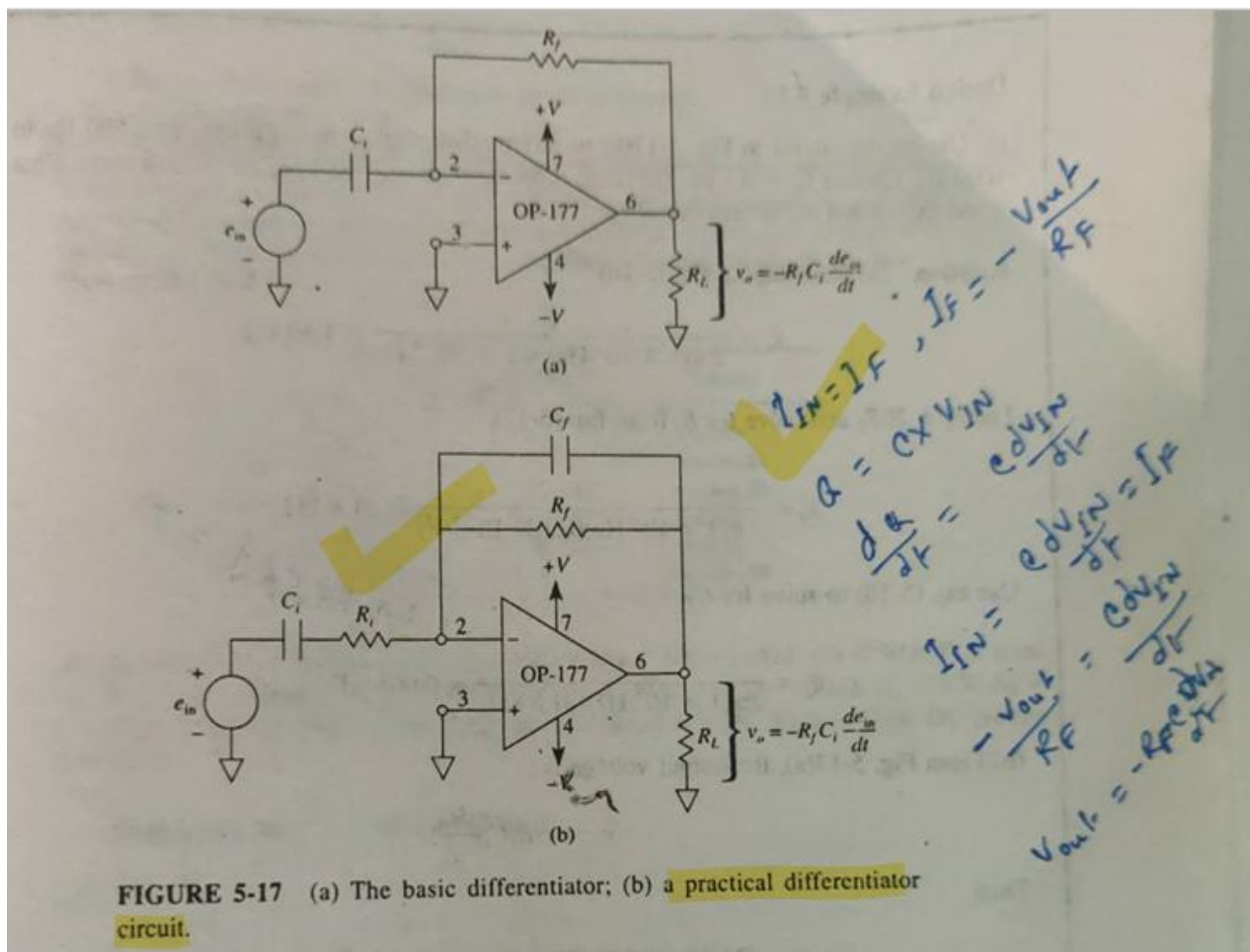
$$v_c = \frac{1}{C} \int i \, dt \quad (5-12)$$

From Chapter 3, we learned that for an inverting circuit the output voltage is negative of the input voltage. Therefore, substituting Eq. (5-11) into Eq. (5-12) and knowing that the output is negative of the input yields

$$v_o = -\frac{1}{R_i C_f} \int e_{in} dt \quad (5-13)$$

Equation (5-13) states that the output voltage is directly proportional to the negative of the integral of the input voltage and indirectly proportional to the time constant $R_i C_f$. If the input wave is a sine wave, the output is a negative cosine wave. The period of the input wave should be greater than the time constant $R_i C_f$.

Explain how an operational amplifier can be used to design Differentiator:



✓ **Working Principle:**

The input signal is applied to the inverting (–) terminal through a capacitor.

A resistor is used in the feedback path instead of a capacitor.

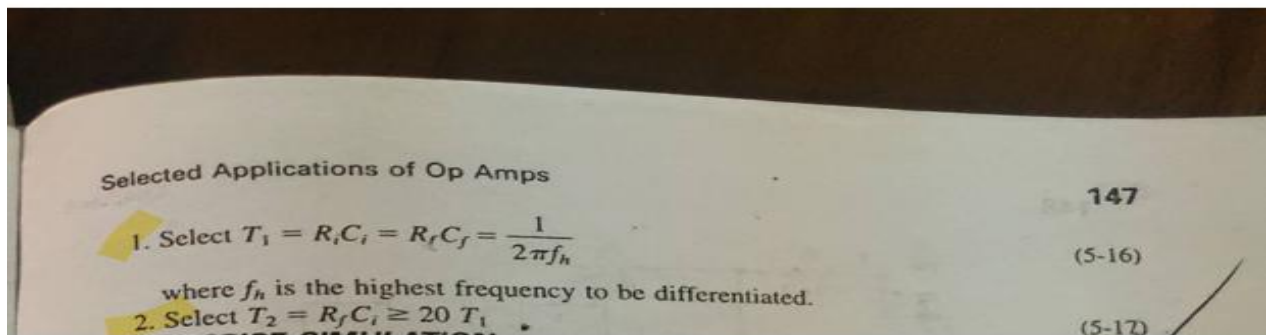
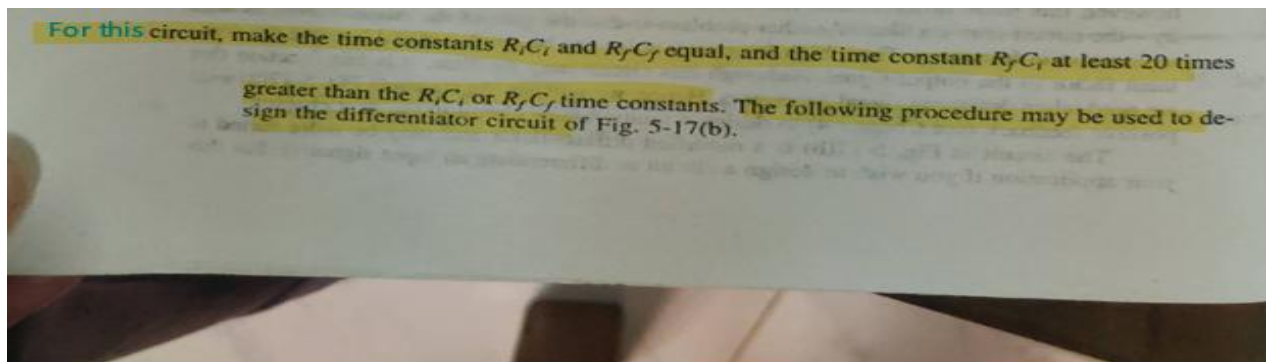
The non-inverting (+) terminal is grounded.

The op-amp responds to the rate of change of the input signal and produces an output accordingly.

✓ **Key Result:**



The output is proportional to the derivative (rate of change) of the input signal and is inverted.



Filters

A **filter** is a circuit that is designed to pass a specified band of frequencies while attenuating all signals outside this band. Filter networks may be either active or passive. **Passive filter networks** contain only resistors, inductors, and capacitors. **Active filters**, which are the only type covered in this text, employ transistors or op amps plus resistors, inductors, and capacitors. Inductors are not often used in active filters, because they are bulky and costly and may have large internal resistive components.

Why filter is used?

Filters are used to selectively pass desired components (frequencies, data, or materials) while blocking, attenuating, or removing unwanted ones, such as noise, distortion, impurities, or irrelevant information.

Show the frequency-response for low, high, band pass filters

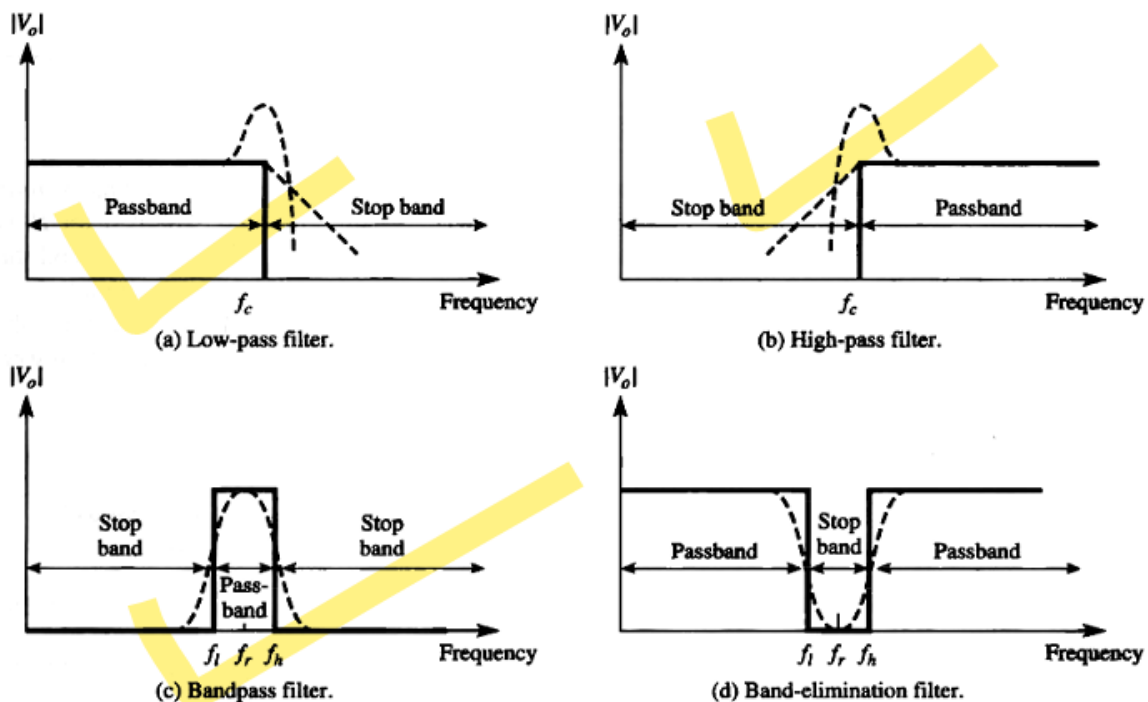


FIGURE 11-1 Frequency response for four categories of filters.

Design low, high, band pass filters

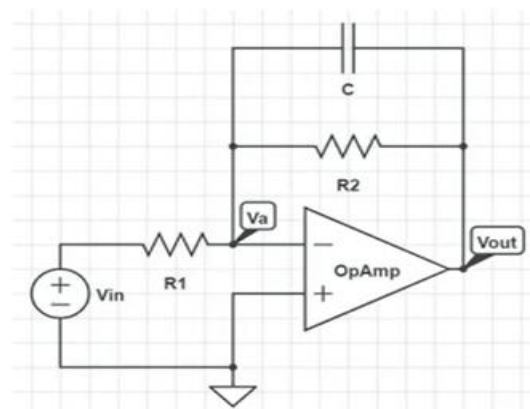
Inverting Low Pass Filter

$$\frac{V_{out}}{V_{in}} = \frac{-Z_2}{Z_1}$$

$$\frac{V_{out}}{V_{in}} = \frac{-Z_2}{Z_1} = \frac{-R_2}{R_1 \times (1 + j2\pi fCR_2)}$$

$$\frac{-R_2}{R_1 \times (1 + j2\pi fCR_2)} = \frac{-R_2}{R_1} \times \frac{1}{1 + j2\pi fCR_2}$$

Plug in $C = 0$.
This turns into $1/1$



age across capacitor C equals output voltage V_o , because this circuit is a voltage follower. E_i divides between R and C . The capacitor voltage equals V_o and is

$$V_o = \frac{1/j\omega C}{R + 1/j\omega C} \times E_i \quad (11-1a)$$

where ω is the frequency of E_i in radians per second ($\omega = 2\pi f$) and j is equal to $\sqrt{-1}$. Rewriting Eq. (11-1a) to obtain the closed-loop voltage gain A_{CL} , we have

$$A_{CL} = \frac{V_o}{E_i} = \frac{1}{1 + j\omega RC} \quad (11-1b)$$

where ω_c is the cutoff frequency in radians per second, f_c is the cutoff frequency in hertz, R is in ohms, and C is in farads. Equation (11-2a) may be rearranged to solve for R :

$$R = \frac{1}{\omega_c C} = \frac{1}{2\pi f_c C} \quad (11-2b)$$

Example 11-1

Let $R = 10 \text{ k}\Omega$ and $C = 0.001 \text{ }\mu\text{F}$ in Fig. 11-2(a); what is the cutoff frequency?

Solution By Eq. (11-2a),

$$\omega_c = \frac{1}{(10 \times 10^3)(0.001 \times 10^{-6})} = 100 \text{ krad/s}$$

or

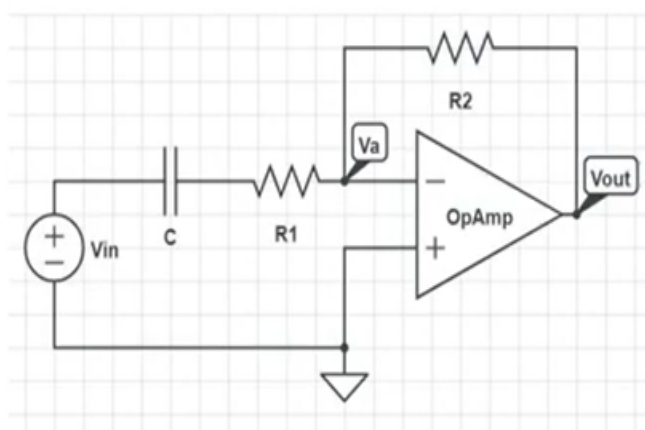
$$f_c = \frac{\omega_c}{6.28} = \frac{100 \times 10^3}{6.28} = 15.9 \text{ kHz}$$

Inverting High Pass Filter

$$\frac{V_{out}}{V_{in}} = \frac{-Z_2}{Z_1}$$

$$\frac{V_{out}}{V_{in}} = H = \frac{-R_2 \times j2\pi fC}{1 + j2\pi fCR_1} \quad \text{For small } C$$

$$f_c = \frac{1}{j2\pi fCR_1} \quad \text{Gain} = \frac{-R_2}{R_1}$$



How D/A converter work?

10.2 DIGITAL-TO-ANALOG CONVERTERS

The input to a D/A converter is an N -bit binary signal, available in parallel form. Normally, digital signals are available at the output of the latches or registers and the voltages, corresponding to logic 0 and logic 1, available to drive the converter are in general not precisely fixed voltages. Therefore, these voltages are not applied directly to the converter but are used to operate digitally controlled switches. The switch is thrown to one of the two positions depending upon the digital signal (1 or 0) which connects precisely fixed voltages $V(1)$ or $V(0)$ to the converter input, corresponding to 1 or 0, respectively.

The analog output voltage V_o of an N -bit straight binary D/A converter is related to the digital input by the equation

$$V_o = K(2^{N-1} b_{N-1} + 2^{N-2} b_{N-2} + \dots + 2^2 b_2 + 2b_1 + b_0) \quad (10.1)$$

where K is a proportionality factor.

$$b_n = \begin{cases} 1 & \text{if the } n\text{th bit of the digital input is 1} \\ 0 & \text{if the } n\text{th bit of the digital input is 0} \end{cases}$$

Example 10.1

Find the analog output voltage of a 4-bit D/A converter for all possible inputs. Assume $K = 1$.

Solution

From Eq. (10.1), we obtain the output voltage for each input and these are given in Table 10.1.

Table 10.1

Digital input				Analog output
b_3	b_2	b_1	b_0	V
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3

0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15

The Analog output voltage = 15 V (maximum value)

5-BIT(b0-b4) /(3-bit(b0-b2) 0-7) HOLE SAME VABE 0-31V AR Analog output voltage = 31 V (maximum value)

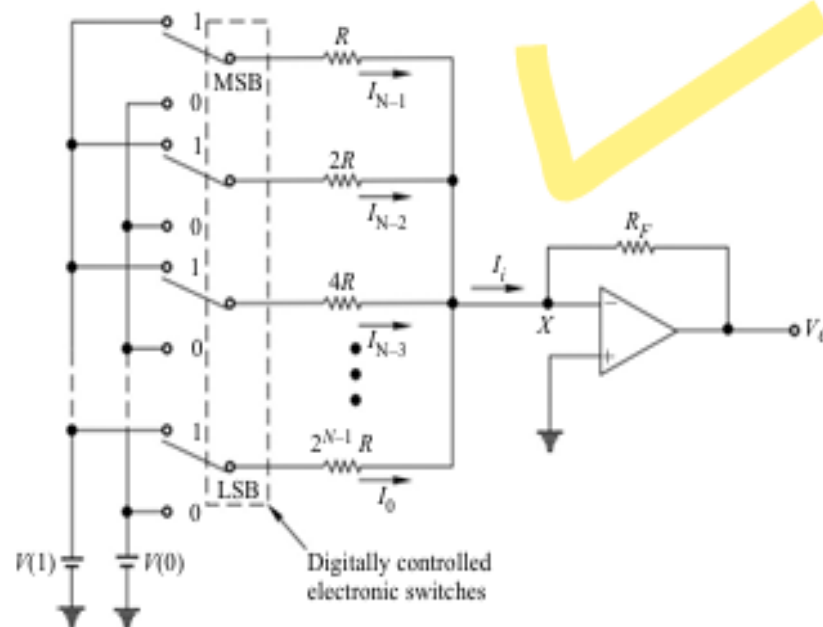


Fig. 10.1 Weighted-Resistor D/A Converter

For straight binary input, $V(0) = 0$ and $V(1) = -V_R$, and the output voltage V_o is given by

$$V_o = -(-V_R) \left(\frac{R_f}{R} b_{N-1} + \frac{R_f}{2R} b_{N-2} + \frac{R_f}{2^2 R} b_{N-3} + \dots + \frac{R_f}{2^{N-1} R} b_0 \right) \quad (10.4)$$

which is of the same form as Eq. (10.1)

with
$$K = \frac{R_f}{2^{N-1} R} \cdot V_R \quad (10.5)$$

The output swings in only one direction and therefore is *unipolar*. If it is required to convert digital data in bipolar format such as in sign-magnitude, 1's complement or 2's complement format, then $V(0) \neq 0$. In such cases, $V(0)$ is used to *offset* the output swing.

With $V(1)$ and $V(0)$ as the voltages applied to the resistor network for 1 and 0, respectively, the output voltage V_o of Fig. 10.1 is given by

$$V_o = \frac{R_f}{2^{N-1} R} (2^{N-1} V_{N-1} + 2^{N-2} V_{N-2} + \dots + 2^1 V_1 + 2^0 V_0) \quad (10.6)$$

An offset can also be produced in the output voltage V_o by using the circuit of Fig. 10.2. The offset voltage produced in this circuit is

$$-\frac{R_f}{R_{\text{off}}} \cdot V_{\text{off}}$$

How offset voltage is used in D/A converter?

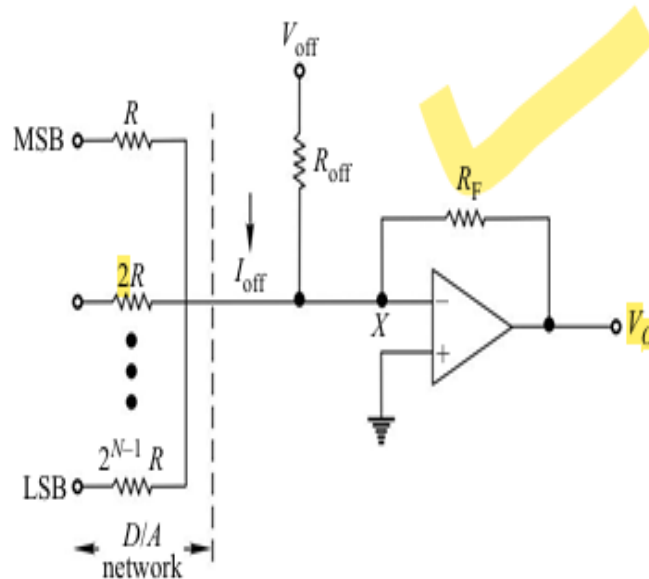


Fig. 10.2 Circuit Used to Offset the Output Voltage

Example 10.2

(a) Consider a 4-bit unipolar D/A converter with $V(1) = -1$ V, $V(0) = 0$ V, and $R_F = 8R$. Obtain the analog output voltage for each of the digital inputs from 0000 to 1111. (b) Adjust the offset voltage, using the circuit of Fig. 10.2, such that $V_o = 0$ V for a digital input of 1000. With this offset, obtain the analog output voltage for each of the digital inputs. (c) With the offset used above, obtain the analog output voltages if the MSB is complemented before applying to the D/A converter.

Solution

(a) Using Eq. (10.6), we obtain the analog voltages which are same as those given in Table 10.1.

10.2.3 Specifications for D/A Converters

The characteristics of a D/A converter, which are generally specified by the manufacturers are:

1. Resolution,
2. Linearity,
3. Accuracy,
4. Settling time, and
5. Temperature sensitivity.

* Why we convert analog to digital signal?

⇒ A When the analog signal is affected by noise, then there's no way to understand it. But for pulse or digital signal, when the signal is affected by noise it can be detected as well as corrected.

◆ Quantization Error

Quantization error is the difference between the actual analog input value and the nearest digital value produced by an A/D (Analog-to-Digital) converter.

In this case, the maximum quantization error for any analog input voltage V_x in the given range is $V/8$.

The quantization error can be reduced if we choose the middle six intervals of size $S = V/7$ and the top and the bottom intervals of size $S/2 = V/14$. The intervals, along with their assigned digital values and equivalent analog output voltage if these digital signals are applied to a D/A converter, are shown in Fig. 10.17. This shows that the maximum quantization error will be $S/2 = V/14$ for any analog input voltage V_x in the range 0 to V . The quantization error can also be specified in terms of LSB. For example, in the above case, the maximum quantization error is $\pm \frac{1}{2}$ LSB.

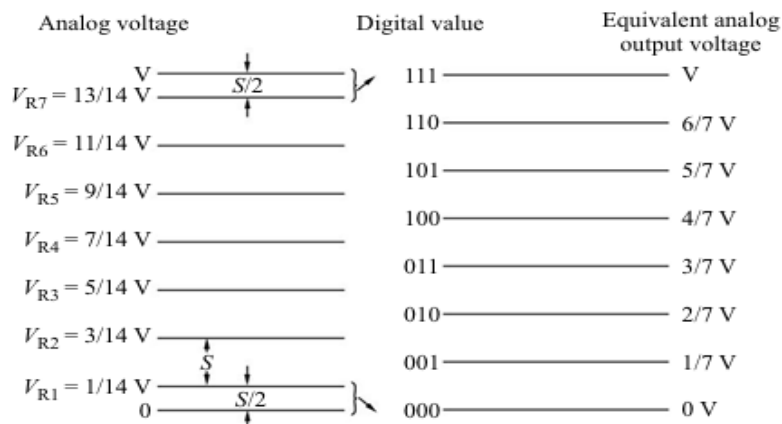


Fig. 10.17 Quantization with Maximum Error = $S/2$

10.5.2 Parallel-Comparator (or Flash) A/D Converter

A 3-bit parallel-comparator A/D converter is shown in Fig. 10.19. V_x is the analog voltage to be converted into digital form. The voltage corresponding to full scale is V from which the reference voltages $V_{R1}, V_{R2}, \dots$ (See Fig. 10.17) are generated using the resistor network. The voltage V_x is compared simultaneously with the reference voltages by using comparators. A 7-bit output is obtained from the comparators which is stored in latches. This 7-bit digital signal is converted to a 3-bit output by using a decoder circuit. The comparator outputs and the 3-bit digital output for each interval of the analog voltage are given in Table 10.6.

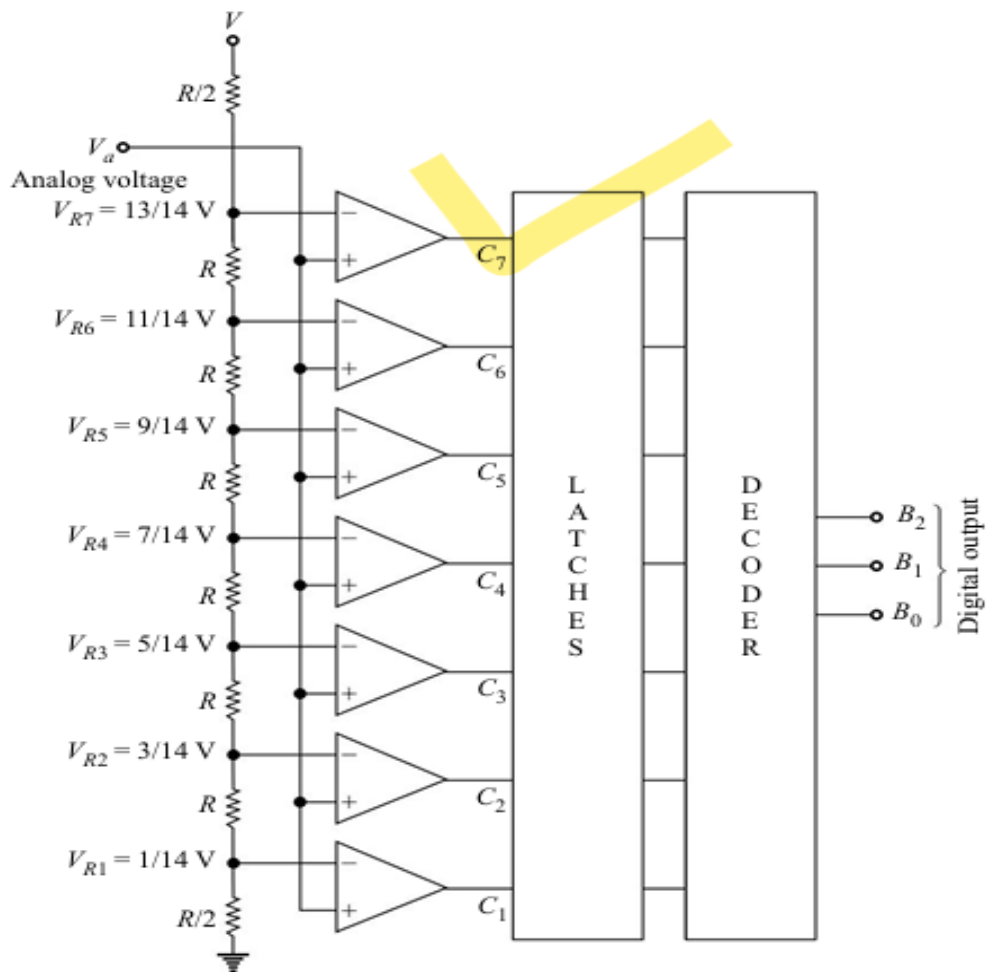


Fig. 10.19 **A 3-bit Parallel-Comparator (Flash) A/D Converter**

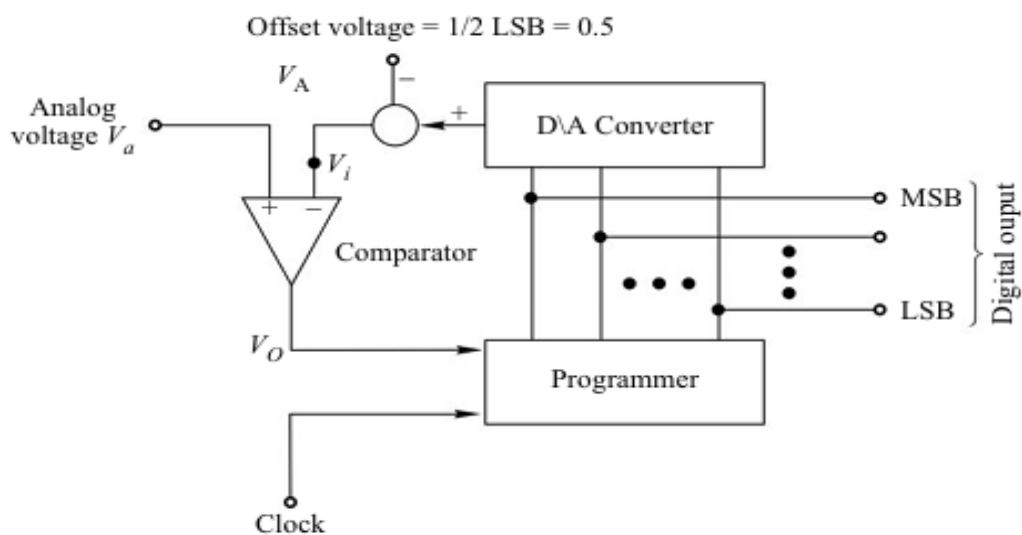


Fig. 10.22 **Successive-Approximation A/D Converter**

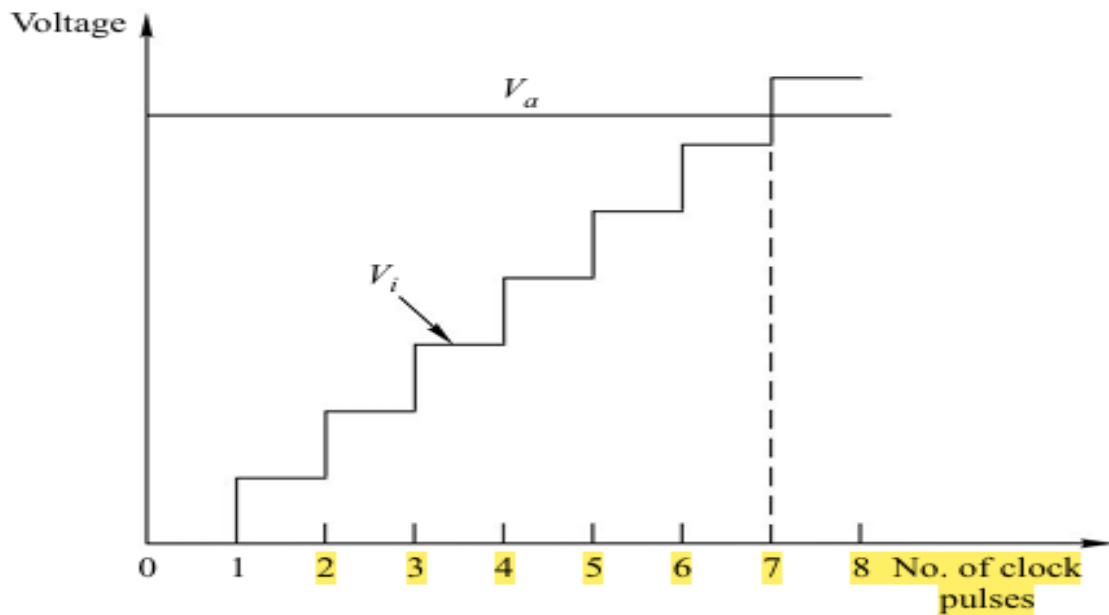


Fig. 10.23 Waveform of Counting A/D Converter

10.5.8 Specifications of A/D Converters

The following specifications are usually specified by the manufacturers of A/D converters:

1. Range of input voltage,
2. Input impedance,
3. Accuracy,
4. Conversion time, and
5. Format of digital output.

Examples of where D/A and A/D converters are used

A D/A converter is used when **digital data needs to be converted into real-world analog signals.**

Examples:

- **Music players / Mobile phones** → digital audio converted to analog sound for speakers
- **TV / Display systems** → digital video signals converted to analog display signals

- **Function generators** → digital control signals converted into analog waveforms
 - **Engine control systems** → digital ECU outputs converted to analog control voltages
-

An A/D converter is used when **real-world analog signals need to be converted into digital form for processing.**

Examples:

- **Microphones** → sound (analog) converted into digital audio in computers
- **Digital cameras** → light (analog) converted into digital image pixels
- **Temperature sensors** → analog temperature converted into digital data
- **Medical equipment (ECG, EEG)** → body signals converted into digital form for analysis
- **Mobile sensors** → acceleration, sound, light converted into digital signals

How 4 bit digital output can be obtained from an analog voltage in the range of 0 to V ?

A 4-bit ADC gives:

$$2^4 = 16 \text{ levels}$$

Step size (resolution):

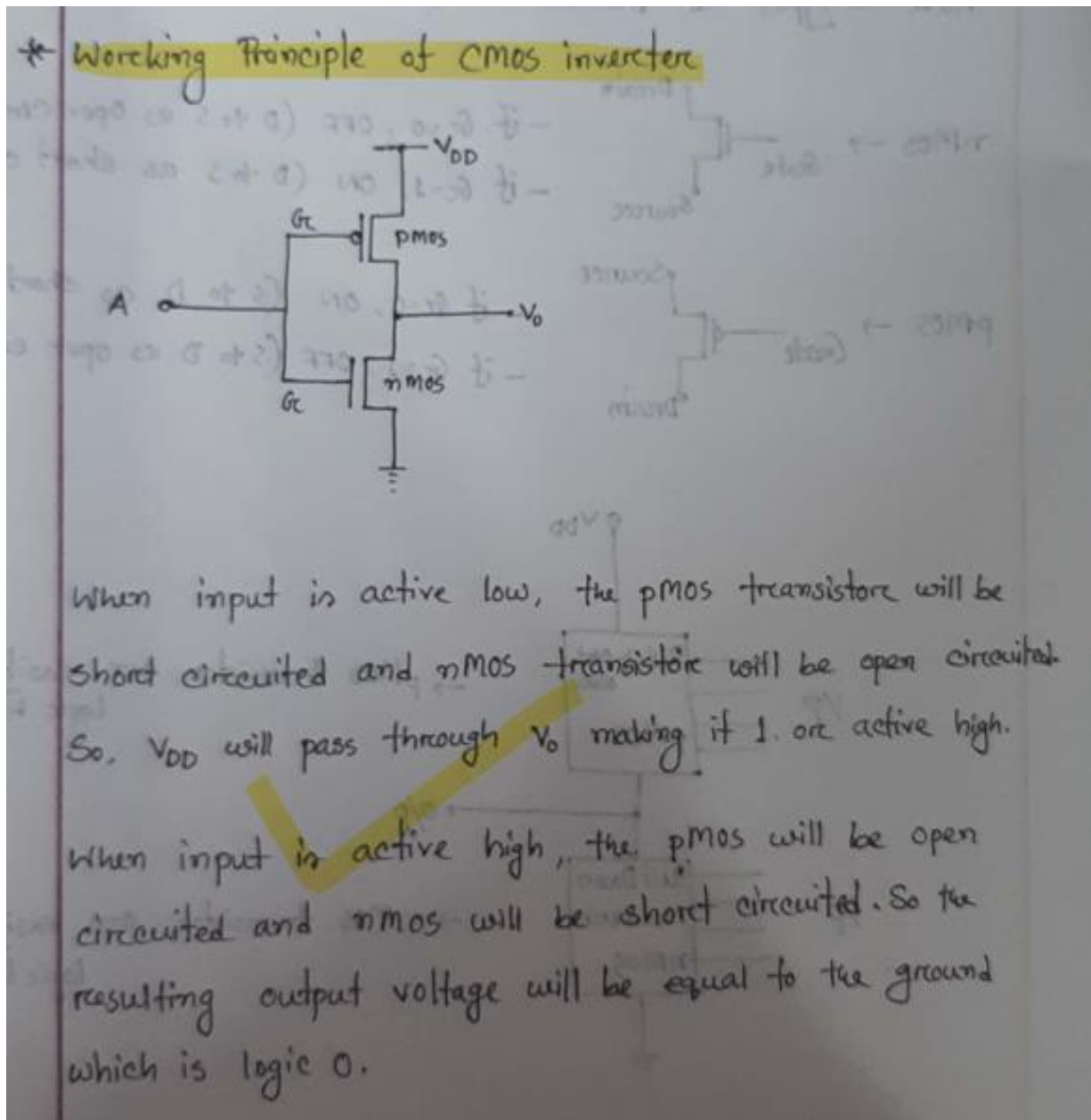
$$\Delta = \frac{V}{16}$$

Digital output code:

$$\text{Code} = \left\lfloor \frac{V_{in}}{\Delta} \right\rfloor = \left\lfloor \frac{16V_{in}}{V} \right\rfloor$$

So:

- $V_{in} = 0 \Rightarrow 0000$
- $V_{in} = V \Rightarrow 1111$



1. What do you mean by cut-off frequency?

Cut-off frequency is the frequency at which the output of a filter drops to **70.7% (or -3 dB point)** of its maximum value.

- Below cut-off frequency $\rightarrow$ signal passes (for low-pass filter)
- Above cut-off frequency $\rightarrow$ signal is blocked

2. Why and how offset voltage is used in D/A converter?

Why it is used:

In practical DACs, output may not start exactly from zero due to errors in components. This causes **offset error**.

How it is used:

- A small DC voltage (offset voltage) is added to correct the output.
- It shifts the output curve so that:
 - Input = 0 → Output = 0 V (ideal condition)

3. Advantages of active filter

Advantages:

- No inductors required (smaller size)
- Provides signal gain
- Easy to design and tune
- High input impedance → no loading effect
- Low output impedance
- Better performance at low frequencies

4. What is operational amplifier (Op-Amp)?

An **operational amplifier (Op-Amp)** is a **high gain voltage amplifier** with:

- Two inputs (inverting and non-inverting)
- One output

It amplifies the difference between two input voltages:

$$V_{out} = A(V^+ - V^-)$$

5. Problems of weighted resistor DAC

- Requires wide range of resistor values (R , $2R$, $4R$, $8R$...)
- Difficult to fabricate accurate resistors
- Not suitable for high-bit resolution (e.g., 8-bit, 12-bit)
- Sensitive to resistor tolerance errors
- Large power consumption for many bits

6. Differences between ideal and practical Op-Amp

Feature	Ideal Op-Amp	Practical Op-Amp
Gain	Infinite	Very high but finite
Input impedance	Infinite	High but finite
Output impedance	Zero	Very low but not zero
Bandwidth	Infinite	Limited
Offset voltage	Zero	Present (small)
Current draw	Zero	Small bias currents
Frequency response	Perfect	Limited